Chapter 10: Understanding Correlation Output

Interpreting Relationships Between Variables

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1 Overview

This reading prepares you for the lab session where you'll explore **correlations**. Correlation tells us whether two variables are related and how strongly. You'll learn to interpret correlation output and create meaningful scatterplots.

> **i What You'll Learn**
>
> By the end of this lab, you'll be able to:
>
> - Interpret correlation coefficients (r values)
> - Understand what scatterplots reveal about relationships
> - Read and interpret correlation test output
> - Understand why correlation does not imply causation

> **Interactive Tool: Correlation Playground**
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> Explore correlation interactively:
>
> 🎮 **Launch Correlation Playground**
>
> Adjust correlation strength with sliders and see how scatterplots change in real-time. Build your intuition for what different r values look like.

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2 What Is Correlation?

Correlation measures the strength and direction of the linear relationship between two continuous variables.

- **Strength**: How tightly the data points cluster around a line (from weak to strong)
- **Direction**: Whether variables increase together (positive) or one increases as the other decreases (negative)

2.1 The Correlation Coefficient (r)

The correlation coefficient **r** ranges from −1 to +1:

Value	Interpretation
r = +1	Perfect positive relationship
r = +0.7	Strong positive relationship
r = +0.3	Weak positive relationship
r = 0	No linear relationship
r = −0.3	Weak negative relationship
r = −0.7	Strong negative relationship
r = −1	Perfect negative relationship

! Direction vs Strength

The **sign** tells you direction; the **absolute value** tells you strength. r = −0.8 is a stronger relationship than r = +0.3.

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3 Reading Scatterplots

3.1 Why Visualise?

Before looking at numbers, **always** examine a scatterplot. It reveals:

- The direction of the relationship (positive, negative, or none)
- The strength of the relationship (tight cluster vs spread out)
- The shape (linear vs curved)
- Any unusual patterns or outliers

3.2 What to Look For

Each point represents one participant:

- **X-axis**: One variable
- **Y-axis**: The other variable

Pattern Guide:

Pattern	Meaning
Upward slope	Positive relationship
Downward slope	Negative relationship
No clear slope	No relationship
Tight cluster around line	Strong relationship
Wide scatter	Weak relationship

3.3 Visual Examples

```
Strong Positive (r ≈ 0.8)    Weak Positive (r ≈ 0.3)    No Relationship (r ≈ 0)
        •                           •  •                      •   •
       • •                        •    •  •                 •     •
      •  ••                      •  •   •                 •  •    •
     •    •                       •  •                       • •  •
    •                            •                        •    •
```

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4 Interpreting Correlation Output

When you run a correlation test, you'll see output like this:

```
    Pearson's product-moment correlation

    data:  reaction_time and time_estimate
    t = 3.12, df = 48, p-value = 0.003
    95 percent confidence interval:
     0.15  0.63
    sample estimates:
          cor
         0.42
```

4.1 What Each Part Means

Output	Meaning
`t = 3.12`	The test statistic
`df = 48`	Degrees of freedom (sample size - 2)
`p-value = 0.003`	Probability of this r if there's truly no relationship
`95 percent confidence interval`	Plausible range for the true correlation
`cor = 0.42`	The sample correlation coefficient

4.2 Key Questions to Answer

1. **What is r?** → 0.42 (medium positive correlation)
2. **Is it significant?** → Yes, $p = 0.003 < 0.05$
3. **What's the CI?** → [0.15, 0.63] - the true r is likely in this range
4. **What does it mean?** → Higher reaction times are associated with higher time estimates

Interpreting the Confidence Interval

A 95% CI of [0.15, 0.63] means we're fairly confident the true population correlation is somewhere in this range. Since it doesn't include 0, the correlation is statistically significant.

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5 Correlation Strength Guidelines

Cohen (1988) suggested these benchmarks:

	r
.10 - .29	Small/weak
.30 - .49	Medium/moderate
.50+	Large/strong

Context Matters

These are rough guidelines. In some research areas, $r = .30$ is typical and meaningful. In others, $r = .50$ might be expected.

5.1 Understanding r-squared

Squaring r tells you the **proportion of variance explained**:

r	r^2	Interpretation
.10	.01	1% shared variance
.30	.09	9% shared variance
.50	.25	25% shared variance
.70	.49	49% shared variance

! Reality Check

A correlation of $r = .42$ sounds reasonable. But $r^2 = .18$ means one variable explains only 18% of the variance in the other. The remaining 82% is due to other factors!

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6 Pearson vs Spearman

6.1 Pearson's r (Most Common)

Use when:

- Both variables are continuous
- Relationship is approximately linear
- Data are roughly normally distributed

6.2 Spearman's rho (ρ)

Use when:

- Data are ordinal (ranked)
- Relationship is monotonic but not linear
- Data have outliers or are skewed

Both are interpreted the same way — they range from -1 to $+1$ with the same strength guidelines.

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7 Correlation Does NOT Mean Causation

⚠ Critical Point

A correlation between X and Y does **NOT** mean X causes Y. This is possibly the most important lesson in research methods.

7.1 Why Not?

Three possible explanations for any correlation:

1. **X causes Y**: More sleep $\longrightarrow$ better performance
2. **Y causes X**: Better performance $\longrightarrow$ more sleep (less stress)
3. **Z causes both**: Good health $\longrightarrow$ more sleep AND better performance

Without experimental manipulation, we cannot determine which explanation is correct.

7.2 Classic Example

Ice cream sales and drowning deaths are correlated.

Does ice cream cause drowning? No! A third variable (hot weather) causes both:

- Hot weather $\longrightarrow$ more ice cream sales
- Hot weather $\longrightarrow$ more swimming $\longrightarrow$ more drowning

7.3 DataLab Example

Self-rating and throwing performance might be correlated.

Does thinking you're good make you perform better? Possible alternatives:

- Past experience → higher self-rating AND better performance
- Natural ability → higher self-rating AND better performance

We can't determine causation from correlation alone.

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8 Reporting Correlations (APA Style)

8.1 Template

> Variable X and Variable Y were [significantly/not significantly] [positively/negatively] correlated, r(df) = .XX, p = .XXX, 95% CI [.XX, .XX].

8.2 Example

> Reaction time and time estimation were significantly positively correlated, $r(48) = .42$, $p = .003$, 95% CI [.15, .63].

8.3 Elements to Include

- The variables being correlated
- Direction (positive/negative)
- Degrees of freedom in parentheses
- The r value
- The p-value
- The 95% CI

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9 Preparation Checklist

Before arriving at the lab:

> ⚠ Before You Arrive
>
> - ☐ Read this chapter carefully
> - ☐ Try the Correlation Playground app
> - ☐ Review the lecture material on correlation
> - ☐ Understand what r values mean (strength and direction)
> - ☐ Be able to interpret example output

9.1 Questions to Consider

- Which DataLab variables might be correlated?
- What would a positive correlation between reaction time and time estimate mean?
- How would you explain correlation vs causation to a friend?

10 Summary

Concept	Key Point
Correlation (r)	Measures strength and direction of linear relationship
Range	−1 (perfect negative) to +1 (perfect positive)
Scatterplot	Always visualise before interpreting numbers
r^2	Proportion of shared variance
Significance	$p < .05$ suggests relationship exists in population
Causation	Correlation does NOT imply causation

11 Correlation Strength Quick Reference

	r		Interpretation
.10	Small	.01	1%
.20	Small-medium	.04	4%
.30	Medium	.09	9%
.40	Medium-large	.16	16%
.50	Large	.25	25%
.70	Very large	.49	49%